

**Project Design Phase-II**  
**Solution Requirements (Functional & Non-functional)**

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| Date          | 25 June 2026                                                      |
| Team ID       | SWTID-2026-8514                                                   |
| Project Name  | Visualization Tool for Electric Vehicle Charge and Range Analysis |
| Maximum Marks | 4 Marks                                                           |

**Functional Requirements:**

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                                |
|--------|-------------------------------|---------------------------------------------------------------------------------------------------|
| FR-1   | Data Collection               | Import Electric Vehicle datasets (EV data, Charging Stations, EV India, Cheapest EV Cars).        |
| FR-2   | Data Processing               | Clean, validate, and organize datasets for analysis.                                              |
| FR-3   | SQL Analysis                  | Execute SQL queries to analyze vehicle range, pricing, charging stations, brands, and efficiency. |
| FR-4   | Dashboard Visualization       | Create interactive Tableau dashboards with charts, maps, filters, and KPIs.                       |
| FR-5   | Story Visualization           | Develop Tableau Story to present insights in a sequential format.                                 |
| FR-6   | Dashboard Integration         | Embed Tableau Dashboard and Story into the Flask web application.                                 |
| FR-7   | User Interaction              | Allow users to explore dashboards using filters, maps, and interactive visualizations.            |
| FR-8   | Project Repository            | Store source code, datasets, SQL scripts, documentation, and project files in GitHub.             |

## Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

| NFR No. | Non-Functional Requirement | Description                                                                     |
|---------|----------------------------|---------------------------------------------------------------------------------|
| NFR-1   | Usability                  | User-friendly interface with interactive dashboards and easy navigation.        |
| NFR-2   | Performance                | Fast loading and smooth visualization of EV datasets.                           |
| NFR-3   | Reliability                | Accurate data processing and consistent dashboard performance.                  |
| NFR-4   | Availability               | Dashboard and project accessible through Tableau Public and GitHub.             |
| NFR-5   | Scalability                | Architecture supports adding new datasets, dashboards, and analytical features. |
| NFR-6   | Maintainability            | Modular project structure for easy updates and future enhancements.             |
| NFR-7   | Security                   | Secure project hosting using GitHub and HTTPS-enabled Tableau Public links.     |